01 — Project Introduction

Project information

Item	Details
Course	DBMS Project — Semester 5
Team	Sneha Angelin Jisho (PES2UG24CS507); Sowmya Ramesh (PES2UG24CS512)
Stack	React + Vite; Node.js + Express; MySQL; MongoDB
Repository	https://github.com/snehajisho/Satellite_Data_Provenance_Earth_Observatory-DBMS_Project

1. Problem Statement

`Satellite Data Provenance and Earth Observations`

Earth-observation missions generate satellite observations and associated information such as mission and satellite identity, sensor characteristics, acquisition time, geographic coverage, data products, quality indicators, image references and processing history. If these records are not connected and searchable, it becomes difficult to retrieve observations, relate products to their source records and understand how a product was produced.

This project develops a web-based repository that combines MySQL for structured relational records and MongoDB for flexible, nested observation metadata. A React interface communicates with a Node.js/Express backend to present satellite information, observations, metadata, quality information and processing provenance.

2. Objectives

- Design a relational schema for missions, satellites, satellite–mission assignments, sensors, ground stations, geographic locations, observations, data products, processing activities and users.
- Store nested observation metadata, spectral information, quality information, processing metadata and image references in MongoDB.
- Build a web interface for exploring satellite, observation, analytics, provenance and quality information.
- Expose REST API endpoints through Node.js and Express.

- Implement user registration and login using bcrypt password hashing and JWT-based authentication.
- Enforce data integrity using keys, constraints, indexes, views, stored procedures, a function and triggers.
- Demonstrate SQL and MongoDB queries and verify the implementation using test suites.

3. Scope

3.1 Included functionality

- Store and retrieve sample mission, satellite, sensor and ground-station records.
- Store geographic areas and observations with time, orbit number, cloud-cover percentage and status.
- Associate observations with data products and processing activities.
- Store nested observation metadata and image paths in MongoDB.
- Retrieve and search metadata by observation, location and satellite.
- Authenticate users and enforce role-based access to protected backend operations.
- Provide Home, Satellites, Observations, Analytics, Provenance and Quality views in the frontend.

3.2 Boundaries and assumptions

The repository uses sample records for academic demonstration. The sample observations and processing activities are not independently verified acquisition/production records. The SQL seed script explicitly describes its ground-station records as fictional sample records, not verified receiving sites. The project is not a full satellite ground segment or a production remote-sensing processing pipeline.

3.3 Intended users

- **USER** accounts can access permitted read operations after authentication.
- **ADMIN** accounts have additional permissions for operations protected by administrator authorization.
- Developers and evaluators can inspect database scripts, API behaviour, frontend functionality and test evidence.